



# More Than One Fix

Ontario Math C3.2 — find a bug, then choose the best fix

Name: \_\_\_\_\_

Date: \_\_\_\_\_

★ I can fix a loop bug more than one way and say which fix I would choose.

## Some bugs have two fixes

Bit's loop does TOO MUCH. The code repeats two actions, but Bit only needs to do the action half as many times.

There are two good ways to fix it: change the REPEAT COUNT, or change the BLOCKS INSIDE the loop. Both reach the goal — find both, then pick one to explain. Tip: total actions = repeat × blocks inside.

## Bit should clap 4 times — but this claps 8

```
repeat 4 {  
  clap  
  clap  
}
```

BLOCKS

### OUTPUT

Each repeat = 2 claps. 4 repeats × 2 = 8 claps. Goal is 4.

### 1 Find both fixes

Bit should clap 6 times, but repeat 6 { clap, clap } claps 12 times. Fix A: change the repeat count to \_\_\_\_\_. Fix B: keep repeat 6 but put \_\_\_\_\_ clap(s) inside. Show each fix gives 6.

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### 2 Hops

repeat 10 { hop, hop } makes 20 hops, but Bit should hop 10 times. Write Fix A (the new repeat count) and Fix B (the blocks inside). Which one changes fewer blocks?

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### 3 Write it

Bit should stomp 8 times. The buggy code is repeat 8 { stomp, stomp } = 16 stomps. Write the WHOLE corrected loop two ways — one using Fix A, one using Fix B.

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### 4 Challenge — choose and justify

Bit should clap 12 times. The buggy code is repeat 6 { clap, clap, clap } = 18 claps. Find a fix that changes the repeat count (Fix A) and a fix that changes the blocks inside (Fix B). Pick the one you would use and say WHY in one sentence.

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## Big idea

A bug can have more than one correct fix. A good coder finds them, then chooses the one that is simplest or clearest — and can say why.



# More Than One Fix

Quick facilitation notes & answer key

## What this worksheet teaches

Students find a loop bug that has TWO good fixes — change the repeat number, or change the blocks inside — show both reach the goal, then choose one and say why. Often there is more than one way to fix a bug. No coding experience needed.

## Before you start

No computer needed — the worksheet shows the code and what Bit should do. Optional: a Scratch-style app (for the block loops) or a free Python-turtle playground (for the shapes) to run the code — you only press Run.

## How to lead it (about 30 minutes)

1. Show it (you do). A loop that does double the goal can be fixed by halving the repeat OR by removing one inside block.
2. Do one together. Exercise 1 — Fix A: repeat 3 with two claps; Fix B: repeat 6 with one clap.
3. Let them work. Students give both fixes (Ex 2), write a whole corrected loop (Ex 3), and choose a fix with a reason (Ex 4).

## Answer key (these are the verified correct answers)

Exercise 1 — Fix A: repeat 3 ( $3 \times 2 = 6$ ). Fix B: one clap inside ( $6 \times 1 = 6$ ).

Exercise 2 — Fix A: repeat 5 ( $5 \times 2 = 10$ ). Fix B: one hop inside ( $10 \times 1 = 10$ ). Fix B changes fewer blocks.

Exercise 3 — Fix A: repeat 4 with two stomps = 8; Fix B: repeat 8 with one stomp = 8. Exercise 4 (challenge) — Fix A: repeat 4 with three claps = 12; Fix B: repeat 6 with two claps = 12. Any reasoned choice is accepted.

## If students get stuck — and ways to adjust

Watch for: thinking there is only one fix. Show that changing the count OR the inside blocks can both work.

Make it easier: check each fix by multiplying repeats  $\times$  blocks inside.

Make it harder: ask which fix changes the fewest blocks and why that can be better.

## Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, find what is wrong, fix it, and explain what the change did.

## You'll know it worked when

The child gives two working fixes, shows both reach the goal, and justifies a choice.